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Automatic discovery and enterprise control of a robotic workforce

Abstract

In an embodiment, a method includes receiving bot presence data from a bot discovery agent executing in a computing environment. The method also includes determining, based on the bot presence data, whether an unauthorized bot is present in the computing environment. The method also includes, responsive to a determination, based on the bot presence data, that an unauthorized bot is present in the computing environment, automatically initiating a remediation workflow relative to the unauthorized bot.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This patent application claims priority to U.S. Provisional Application No. 63/272,028 filed on Oct. 26, 2021. U.S. Provisional Application No. 63/272,028 filed on Oct. 26, 2021 is hereby incorporated by reference.

BACKGROUND

Technical Field

(1) The present disclosure relates generally to robotic processes and more particularly, but not by way of limitation, to automatic enterprise control of unauthorized robotic workforce.

History of Related Art

(2) Task automation, while often desirable, is often performed in an uncontrolled and generally haphazard manner. Users generally implement automation in any way that seems appropriate to them without awareness of organizational policies and risk.

SUMMARY

(3) A system of one or more computers can be configured to perform particular operations or actions by virtue of having software, firmware, hardware, or a combination of them installed on the system that in operation causes or cause the system to perform the actions. One or more computer programs can be configured to perform particular operations or actions by virtue of including instructions that, when executed by data processing apparatus, cause the apparatus to perform the actions.

(4) In an embodiment, one general aspect includes a method. The method includes receiving bot presence data from a bot discovery agent executing in a computing environment. The method also includes determining, based on the bot presence data, whether an unauthorized bot is present in the computing environment. The method also includes, responsive to a determination, based on the bot presence data, that an unauthorized bot is present in the computing environment, automatically initiating a remediation workflow relative to the unauthorized bot. Other embodiments of this aspect include corresponding computer systems, apparatus, and computer programs recorded on one or more computer storage devices, each configured to perform the actions of the methods.

(5) In an embodiment, another general aspect includes a computer system having a processor and memory. The processor and the memory in combination are operable to implement a method. The method includes receiving bot presence data from a bot discovery agent executing in a computing environment. The method also includes determining, based on the bot presence data, whether an unauthorized bot is present in the computing environment. The method also includes, responsive to a determination, based on the bot presence data, that an unauthorized bot is present in the computing environment, automatically initiating a remediation workflow relative to the unauthorized bot.

(6) In an embodiment, in another general aspect, a computer-program product includes a non-transitory computer-usable medium having computer-readable program code embodied therein. The computer-readable program code is adapted to be executed to implement a method. The method includes receiving bot presence data from a bot discovery agent executing in a computing environment. The method also includes determining, based on the bot presence data, whether an unauthorized bot is present in the computing environment. The method also includes, responsive to a determination, based on the bot presence data, that an unauthorized bot is present in the computing environment, automatically initiating a remediation workflow relative to the unauthorized bot.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) A more complete understanding of the method and apparatus of the present disclosure may be obtained by reference to the following Detailed Description when taken in conjunction with the accompanying Drawings wherein:
- (2) FIG. 1 illustrates an example of a system for implementing a central management system;
- (3) FIG. 2 illustrates an example of an environment for bot discovery and enforcement;
- (4) FIG. 3 illustrates an example of a process for discovering and implementing enterprise control of bots,
- (5) FIG. 4 illustrates an example of a process for determining whether unauthorized bots are present or executing in a computing environment;
- (6) FIG. 5 illustrates an example dashboard; and
- (7) FIG. 6 illustrates an example of a computer system.

DETAILED DESCRIPTION

(8) Robotic process automation (RPA) is the use of software, typically in combination with artificial intelligence (AI) and machine learning capabilities, to handle high-volume sequences of repeatable tasks that previously required humans to perform. These tasks can include interacting with a user interface (UI), for example, to perform, queries, calculations, maintenance of records and transactions, web automations, remote operations, database functions, terminal connections, desktop operations, text operations, mouse simulations, keyboard simulations, folder operations, file handlers, clipboard handlers, combinations of the foregoing and/or the like. RPA technology can mimic a human worker, for example, by logging into applications, entering data, calculating and completing tasks, and logging out.

(9) RPA technology is not always part of an organization's information technology (IT) infrastructure, but can instead, sit on top of it, enabling an organization to implement RPA technology quickly and efficiently, without changing the existing infrastructure and systems. In various embodiments, a computing environment can include, for example, many hundreds or thousands of software agents or macros, often referred to herein as "bots," that automatically perform robotic processes. In various cases, the sequence of computer-implemented tasks performed by each bot can be the same or different, or a combination thereof. Oftentimes, the computing environment can further include, for example, many hundreds or thousands of human workers that likewise execute the same or similar computer-implemented tasks, or a combination thereof, via manual interaction with UIs. A set or sequence of computer-implemented tasks, such as the foregoing example tasks, may be referred to as a user-executed process.

(10) Although bots, potentially in combination with human workers, can provide greater efficiency via increased automation, bots introduce numerous technical problems. For example, in many organizations, bot deployments, such as RPA, macros and other forms of automations, are implemented as projects, proof of concepts (POCs) and centers of excellence (COEs) by operations groups, and not in a centralized manner. Many bots function as desktop level RPAs working on users' equipment without server hosting while other bots are hosted by third-party vendors. Employees are authorized to use enterprise assets for their business operations, which involve critical data exchange between various business functions. Business operations and the asset usage are regulated and governed by enterprise compliance & information security groups.

Problematically, various departments and groups in an organization procure hardware and install it in their servers, with or without chief information officer (CIO) or other appropriate approval. At times, organizations acquire hardware and digital assets without documentation, which results in non-compliant and insecure use of hardware and digital assets.

(11) The present disclosure describes examples of automatically discovering previously unknown

bots in an organization and implementing enterprise control of such bots. In various embodiments, discovery agents can be deployed across some or all computing environments in an organization. Thereafter, the bot discovery agents can be leveraged to discover bots and automatically determine, in a centralized manner, whether the bots are authorized or unauthorized. In various embodiments, remediation workflows can be automatically initiated relative to unauthorized bots. Further, in various embodiments, authorized bots and/or unauthorized bots can be automatically subjected to centralized enterprise control. Examples will be described below relative to the Figures.

(12) For purposes of this application, the term “UI” can include, without limitation, any type of human-machine interface. Example UIs include text-based interfaces such as command-line interfaces, conversational interfaces, crossing-based interfaces, direct manipulation interfaces, gesture interfaces, graphical UIs (GUIs), hardware interfaces, holographic UIs, intelligent UIs, motion tracking interfaces, natural-language interfaces, object-oriented UIs, reflexive UIs, touchscreen interfaces, touch UIs, voice UIs, and web-based UIs. Many of the aforementioned UIs accept input via input devices such as a keyboard, mouse, microphone, combinations of the foregoing, and/or the like. Inputs from any of these devices, in possible combination with specific UI objects or components, can be considered examples of UI events. A user, unless specifically stated otherwise, or otherwise understood within the context as used, can include any entity, including a human or bot, that can interact with a UI.

(13) FIG. 1 illustrates an example of a system **100** for implementing a central management system **140** for bot discovery and enforcement. The system **100** includes the central management system **140**, tenant systems **110**, RPA platforms **124**, user systems **160** and one or more data stores **150**, each of which is operable to communicate over a network **108**. The network **108** may be, or include, one or more of a private network, a public network, a local or wide area network, a portion of the Internet, combinations of the same, and/or the like.

(14) In some aspects, the central management system **140** can centrally manage bot deployments on the RPA platforms **124** for its tenants. The RPA platforms **124** are shown to include an RPA platform **124(1)**, an RPA platform **124(2)** and an RPA platform **124(N)**. It should be appreciated that three RPA platforms are shown among the RPA platforms **124** only for illustrative purposes. In various implementations, any number of RPA platforms can be included among the RPA platforms **124**. In a typical embodiment, the RPA platforms **124** each provide RPA software for creating and executing bots, generally using different RPA technology, interfaces and formats.

(15) In particular, in the system **100**, the tenant systems **110** can be served by the central management system **140**. In general, the tenant systems **110** can each be considered an abstraction of users, inclusive of both bots and human workers, that execute user-executed processes managed by the central management system **140**, and the systems and data sources with which those users interact. For example, one of the tenant systems **110** is shown as being owned or operated by “Tenant A” while another system **110** is shown as being owned or operated by a different tenant, “Tenant B.” The tenant systems **110** shown can be owned or operated by the same or different entities. For example, Tenants A and B can represent customers (e.g., entities such as companies or individuals) of an operator of the central management system **140**. Although the term “tenant” is used herein to describe the tenant systems **110** or owners/operators thereof, in addition to having its ordinary meaning, the term “tenant” can, but need not, refer to tenancy in a multitenant software architecture.

(16) The tenant systems **110** are each shown to include one or more managed bots **120**, one or more managed human workers **123**, one or more computer systems **122** and one or more data sources **121**. The one or more computer systems **122** can each provide a computing environment, inclusive of applications and corresponding UIs and dashboards, for executing configurable tasks. As illustrated, any given one of the computer systems **122** may be operated by one of the managed bots **120** or one of the human workers **123**. In some cases, the computer systems **122** may represent desktop virtualization environments. In such cases, the human workers **123**, for example, may

operate the user systems **160** and access the desktop virtualization environments over the network **108**.

(17) The one or more data sources **121** of each of the tenant systems **110** can include data streams or datasets that can be received or processed by the computer systems **122**, potentially as part of executing the configurable tasks. In various cases, the one or more data sources **121** can be updated by the computer systems **122**, or other components, in real-time, on a periodic basis, e.g., according to a schedule, on-demand or a combination of the same. In various cases, the tasks for which the computer systems **122** are configured can be executed in the respective computing environments, for example, manually by the human workers **123** and/or robotically by the managed bots **120**. In a typical embodiment, the managed bots **120** are each deployed on an RPA platform of the RPA platforms **124** and interact as users in the computing environments of the computer systems **122** for purposes executing some or all of the configurable tasks as robotic processes.

(18) The managed bots **120** can each include an instruction set **126** that can be used to provide the managed bots **120** a complete set of all computer-implemented tasks or steps that are to be performed in sequence automatically by the managed bots **120**, for example, as a user-executed process. In some embodiments, the instruction set **126** is in a machine-readable code that can be recognized and executed by a central processing unit. In various embodiments, the instruction set **126** can be made up of, for example, a markup language, a low-level programming language, a high-level programming language, a scripting language, a machine language, an assembly language or any combination thereof.

(19) As illustrated, the computer systems **122** and/or the user systems **160** can include bot discovery agents **130**. The bot discovery agents **130** are software agents that can be used to discover and detect bots that are not centrally managed by the central management system **140** and therefore are not among the managed bots **120**. In certain embodiments, the bot discovery agents **130** can be individually triggered to monitor and record certain information and activity in the computing environments provided by the computer systems **122**. The activity can represent, for example, activity performed by bots. The activity can include, for example, UI activity.

(20) In the illustrated embodiment, the central management system **140** can include a bot enforcer **142**, a bot manager **145**, a bot deployment module **144**, a reporting module **146** and a bot creation module **143**. Each of these components can be implemented with hardware and/or software, including (optionally) virtual machines and containers. In an example, the central management system **140** can be implemented as a single management server. In another example, the central management system **140** can be implemented in a plurality of virtual or physical servers, which may or may not be geographically co-located. In some embodiments, the central management system **140** and/or other aspects of the system **100** may be hosted on a cloud-provider system.

(21) In certain embodiments, features of the components of the central management system **140** can be made accessible over an interface to the user systems **160**. The user systems **160** can include any type of computing device, including desktops, laptops, tablets, and smartphones, to name a few. The user systems **160** can be operated by users, such as the human workers **123**, or by other users, for example, for administration purposes.

(22) The bot enforcer **142** can, for each of the tenant systems **110**, in collaboration with the bot discovery agents **130**, enable automated detection and control of unauthorized bots in any computing environment subject to the central management system **140**. In certain embodiments, the bot enforcer **142** can deploy the bot discovery agents **130** and trigger bot discovery by each the bot discovery agents **130**, or any configurable subset thereof (e.g., by tenant, business unit, user groups, etc.). The bot discovery agents **130** can generate bot presence data, which data indicates an existence of bots in their respective computing environments. The bot discovery agents **130** can send the bot presence data to the bot enforcer **142** for processing. The bot enforcer **142**, in turn, can determine, based on the bot presence data, whether unauthorized bots are executing in the respective computing environments. In various embodiments, a remediation workflow can be

automatically initiated when unauthorized bots are detected. Example implementations of the bot enforcer **142** and/or the bot discovery agents **130** will be described in greater detail relative to FIG. 2.

(23) The bot creation module **143** can be utilized to create bots such as, for example, the managed bots **120**. In some embodiments, an instruction set is created via the bot creation module **143** that contains all of the instructions and/or modules for a specific type of bot. The instruction set that is created can be similar to the instruction set **126** of each of the managed bots **120**. In certain embodiments, the bot creation module **143** can utilize information from the bot enforcer **142** to automatically create each instruction set. In some embodiments, the bot creation module **143** includes or provides a configuration interface for automatic and/or manual creation of an instruction set, or for revision or tuning of an instruction set. The configuration interface can be accessible, for example, by the user systems **160**.

(24) In certain embodiments, the bot deployment module **144** can be utilized to deploy a bot on the RPA platforms **124** for a tenant, such that the deployed bot becomes one of the managed bots **120**. In various embodiments, the bot deployment module **144** can utilize an instruction set created by the bot creation module **143** that can then be deployed. For example, the instruction set **126** of one of the managed bots **120** could be representative of a bot created by the bot creation module **143** and then deployed via the bot deployment module **144**. In certain embodiments, the bot deployment module **144** can be used to deploy a single bot and/or a plurality of bots concurrently. In many cases, the bot deployment module **144** can be utilized to deploy bots on a variety of the RPA platforms **124**. In some embodiments, the bot deployment module **144** can publish a configuration interface to the user systems **160**, for example, for administrators, super users or other users (e.g., of a particular tenant) to select or specify such commands. In other embodiments, the bot deployment module **144** executes an automatic process to deploys bots without manual interaction. In various embodiments, the bot deployment module **144** can store command sets for some or all of the RPA platforms **124**.

(25) The bot manager **145** can serve to manage bots such as, for example, the managed bots **120**, for tenants. In certain embodiments, the bot manager **145** can issue commands to control operation of bots. The bot manager **145** can be utilized to re-configure, optimize and/or customize any of the managed bots **120**. For example, various commands can start or stop bots, perform configuration management, combinations of the same and/or the like. In some cases, the bot manager **145** can publish a configuration interface to the user systems **160**, for example, for administrators, super users or other users (e.g., of a particular tenant) to select or specify such commands.

(26) The reporting module **146** can provide an interface for regular or on-demand reports related to the managed bots **120**, the managed human workers **123**, and/or operation of the bot enforcer **242**. The reporting module **146** can publish reports or other generated information, for example, to a webpage, user dashboard, and/or the like. The reporting module **146** can generate and execute a query of the one or more data stores **150**, as appropriate. The web page, user dashboard or other UI(s) output, for example, by the reporting module **146**, can be accessed by certain users of the user systems **160**.

(27) In general, the one or more data stores **150** can include any information collected, stored or used by the central management system **140**. For example, in various embodiments, the one or more data stores **150** can include instruction sets for bots, command sets for one or more of the RPA platforms **124**, deployment settings for bots (e.g., on a tenant-specific and/or bot-specific basis), metadata for bots to be deployed, data collected from the managed bots **120**, the human workers **123** or the computer systems **122**, combinations of the same and/or the like. In certain embodiments, data stored in the one or more data stores **150** can take the form of repositories, flat files, databases, etc. In certain embodiments, the one or more data stores **150** can be utilized as an event library, in which actions performed by any of the managed bots **120** and/or the user systems **160** are stored. The event library can also contain business rule libraries, exception libraries,

control libraries, keyboard libraries, database libraries and/or cognitive libraries.

(28) FIG. 2 illustrates an example of an environment **200** for a bot enforcer **242**. The environment **200** includes the bot enforcer **242**, bot discovery agents **230**, enforcement settings **252**, internal data sources **254**, and external data sources **256**. In various embodiments, functionality can be distributed between the bot discovery agents **130** and the bot enforcer **142** in any suitable fashion. In general, the bot discovery agents **230** and the bot enforcer **242** can each operate as described relative to the bot discovery agents **130** and the bot enforcer **142**, respectively, of FIG. 1.

(29) The enforcement settings **252** can include, for example, detection settings, authorization settings, and remediation workflow settings. The detection settings can include settings utilized by the bot discovery agents **230** to detect bots in their respective computing environments. For example, the detection settings can indicate what information, file paths, behaviors and/or the like are monitored for and detected by the bot discovery agents **230**, what information is generated, for example, as bot presence data, a frequency of detection, etc. The detection settings can differ, for example, by tenant, organization, customer account, business unit, user group, combinations of the foregoing and/or the like.

(30) The authorization settings in the enforcement settings **252** can include information or data usable to determine, based on bot presence data, whether an unauthorized bot is present or executing in a given computing environment. In certain embodiments, the authorization settings can include signatures of authorized bots, signatures of unauthorized bots, or the like. For any given bot, the signatures can specify, as applicable, applications, observed application behaviors or actions, an RPA platform on which a given bot is deployed, application runtime details, or the like. In addition, or alternatively, the authorization settings can include or define heuristics that are usable to determine, based on the bot presence data, whether an unauthorized bot is present or executing in the given computing environment.

(31) The remediation workflow settings in the enforcement settings **252** can define or establish remediation workflows that are initiated in response to a determination that an unauthorized bot is present or executing in a given computing environment. Each remediation workflow can include one or more actions that are taken. Example actions that may be taken include blocking an unauthorized bot from executing, triggering uninstallation of the unauthorized bot, providing a recommendation to convert an unauthorized bot to an authorized bot, and identifying and flagging the authorized bot to appropriate personnel via a dashboard, notification or other suitable method (e.g., via the reporting module **146**).

(32) The bot discovery agents **230** can initiate a bot discovery process and generate bot presence data based thereon. The bot discovery process can be executed on a continuous or periodic basis in accordance with detection settings of the type described above. In various embodiments, the bot discovery process can include software tracing and recording, file extractions, process monitoring, etc. The bot presence data can include an information set for each bot or potential bot that is detected. For example, the bot presence data can include application or macro names or identifiers, application runtimes, an identification of RPA platforms on which a bot is deployed, observed application behaviors, bot instruction sets similar to the instruction set **126** of FIG. 1, application files, combinations of the foregoing and/or the like. In some cases, the bot presence data can include, for each bot or potential bot, information of a same or similar format as signatures stored in the enforcement settings **252**.

(33) The bot enforcer **242** is operable to deploy the bot discovery agents **230** in computing environments provided by any given system or device managed by the central management system **140** of FIG. 1 such as, for example, the computer systems **122** and/or the user systems **160** of FIG. 1. The bot enforcer **242** can also provide or send, to each of the bot discovery agents **230**, individualized detection settings from the enforcement settings **252**. Such detection settings can be provided initially upon deployment and can be updated on a continuous or ongoing basis.

(34) In various embodiments, the bot enforcer **242** can trigger, or cause, bot discovery by the bot

discovery agents **230** in accordance with their individual detection settings. In various embodiments, the bot enforcer **242** can trigger or cause bot discovery in numerous fashions. In an example, the bot enforcer **242** can trigger or cause bot discovery by sending a command to execute the bot discovery immediately upon receipt of the command. In another example, the bot enforcer **242** can send a command to schedule the bot discovery. In another example, the bot enforcer can trigger or cause bot discovery by virtue of the bot discovery agents **230** having been deployed, such that each of some of the bot discovery agents **230** continuously or periodically execute bot discovery in accordance with their individual detection settings.

(35) Upon receipt of bot presence data from any of the bot discovery agents **230**, the bot enforcer **242** can determine, according to the authorization settings in the enforcement settings **252**, whether an unauthorized bot is present in a computing environment. In various embodiments, the determination of whether an unauthorized bot is present can include applying heuristics and/or mapping the bot presence data to signatures in the enforcement settings **252**. Examples will be described below.

(36) For example, with respect to heuristics, the bot enforcer **242** can, for each bot or potential bot, apply behavior classification and/or general classification of bot presence data. The heuristics can involve, for example, matching the bot presence data for a given bot to an identified authorized bot, to an identified unauthorized bot, or otherwise yielding an authorized or unauthorized determination. In some embodiments, classification can be performed via a supervised machine learning algorithm that predicts authorization or non-authorization based on the bot presence data. In addition, or alternatively, various embodiments can utilize at least one of the classification types such as binary classification, multi-class classification and multi-label classification. In addition, or alternatively, various embodiments can use imbalanced classification, which classification includes techniques such as classification logistic regression, stochastic gradient descent, k-nearest neighbors, decision trees, random forest, artificial neural network, support vector machine and naive bayes.

(37) For example, with respect to signature mapping, the determination of whether an unauthorized bot is present can include mapping the bot presence data to signatures in the enforcement settings **252**. In some cases, the signatures in the enforcement settings **252**, if any, can be indicative of unauthorized bots. In these cases, a determination that the bot presence data for a given bot matches a signature of an unauthorized bot would be indicative that the given bot is unauthorized. In addition, or alternatively, some or all signatures in the enforcement settings **252**, if any, can be indicative of authorized bots. In these cases, a determination that the bot presence data for a given bot matches a signature of an authorized bot would be indicative that the given bot is authorized.

(38) In some embodiments, for any given authorized bot, the bot enforcer **242** can initiate centralized management of the authorized bot such that, with reference to FIG. **1**, the authorized bot may be treated and managed as one of the managed bots **120**. In some embodiments, for any given unauthorized bot, the bot enforcer **242** can initiate, for example, a remediation workflow in accordance with the remediation workflow settings in the enforcement settings **252**. As described previously, various remediation workflows can include, for example, blocking an unauthorized bot from executing, triggering uninstallation of the unauthorized bot, providing a recommendation to convert an unauthorized bot to an authorized bot, and identifying and flagging the authorized bot to appropriate personnel via a dashboard, notification or other suitable method (e.g., via the reporting module **146**). In some cases, the initiation of the remediation workflow by the bot enforcer **242** can include automatically comparing the bot presence data of the unauthorized bot to a signature or other information that resulted in the unauthorized determination, and then automatically identifying one or more specific deficiencies that resulted in the unauthorized determination of that bot.

(39) In an example, if the unauthorized bot is deemed unauthorized because it is deployed using a nonapproved RPA platform, a recommendation or correction may be to deploy the bot on a

different platform (e.g., using one or more approved RPA platforms). In some embodiments involving this scenario, with reference to FIG. 1, the remediation workflow can involve the bot enforcer **242** integrating with the bot creation module **143**, the bot deployment module **144** and the bot manager **145**, as applicable, to stop the unauthorized bot and redeploy and start it on a specified approved RPA platform of the RPA platforms **124**.

(40) In another example, if the unauthorized bot is deemed unauthorized because it includes an unauthorized instruction in an instruction set similar to the instruction set **126** of FIG. 1, a recommendation or correction may be to remove the instruction and/or include an alternative instruction. In some embodiments involving this scenario, with reference to FIG. 1, the remediation workflow can involve the bot enforcer **242** integrating with the bot creation module **143**, the bot deployment module **144** and the bot manager **145**, as applicable, to stop the unauthorized bot, modify the instruction set, and redeploy and start the bot with its modified instruction set.

(41) In various embodiments, the bot enforcer **242** can incrementally or continuously receive new settings for any of the enforcement settings **252**. For example, in some embodiments, new settings can be received from the internal data sources **254**. The internal data sources **254** can include, for example, internal sources or collections of signatures of the type described above. In another example, in some embodiments, new settings can be received from the external data sources **256**. The external data sources **256** can include, for example, settings such as signatures provided by RPA platforms of the RPA platforms **124** of FIG. 1, Internet sources and/or the like.

(42) In various embodiments, the bot enforcer **242** can incrementally or continuously learn new settings for the enforcement settings **252**. In various embodiments, the bot enforcer **242** can initially create one or more machine-learning models for authorized and/or unauthorized bots based on an initial dataset, where each machine-learning model can correspond to a specific robotic process, a step or task of a robotic process, a particular tenant, etc. In some cases, the bot enforcer **242** can create a machine-learning model that is generic, for example, for robotic processes, steps or tasks, and/or tenant. Thereafter, the bot enforcer **242** can facilitate incremental learning based on new information. Models created and updated by the bot enforcer **242** can be included in the enforcement settings **252** or other memory.

(43) FIG. 3 illustrates an example of a process **300** for discovering and implementing enterprise control of bots. In certain embodiments, the process **300** can be implemented by any system that can process data. Although any number of systems, in whole or in part, can implement the process **300**, to simplify discussion, the process **300** will be described in relation to particular components shown and described relative to FIGS. 1-2.

(44) At block **302**, the bot enforcer **242** triggers bot discovery for one or more discovery agents of the bot discovery agents **230**. In certain embodiments, the bot enforcer **242** can trigger bot discovery in any of the ways described above relative to FIG. 2. In various embodiments, the bot enforcer **242** can trigger bot discovery by a single bot discovery agent of the bot discovery agents or by any suitable aggregate of the same. For simplicity of description, the process **300** will be described relative to a single triggered discovery agent of the bot discovery agents **230**.

(45) At block **304**, the triggered discovery agent initiates a bot discovery process in its computing environment as described relative to FIG. 2. At block **306**, the triggered discovery agent generates bot presence data as described relative to FIG. 2. At block **308**, the triggered discovery agent sends the bot presence data to the bot enforcer **242**. At block **310**, the bot enforcer **242** receives the bot presence data.

(46) At decision block **312**, the bot enforcer **242** determines, based on the bot presence data, whether an unauthorized bot is present or executing in the computing environment. In various embodiments, the determination at the decision block **312** can involve any of the methodologies described relative to FIG. 2. If it is determined at the decision block **312** that no unauthorized bot has been detected, the process **300** ends. Otherwise, if it is determined at the decision block **312** that an unauthorized bot is present or executing in the computing environment, the process **300** can

process to block **314**. At block **314**, the bot enforcer **242** automatically initiates a remediation workflow as described relative to FIG. **2**. After block **314**, the process **300** ends.

(47) FIG. **4** illustrates an example of a process **400** for determining whether unauthorized bots are present or executing in a computing environment. The process **400** is a logical view of example functionality that may be performed relative to the computing environment, for example, as all or part of blocks **306-312** of the process **300** of FIG. **3**. In certain embodiments, the process **400** can be implemented by any system that can process data. Although any number of systems, in whole or in part, can implement the process **400**, to simplify discussion, the process **300** will be described in relation to particular components shown and described relative to FIGS. **1-2**.

(48) At block **402**, a triggered discovery agent, from the bot discovery agents **230**, and/or the bot enforcer **242**, perform a first process iteration that includes developing process data. The process data can include, for example, a list of processes executing in the computing environment. For example, in some embodiments, the process data can include information like the information listed below in Table 1.

(49) TABLE-US-00001 TABLE 1 PROCESS NAME TaniumClient iisexpress chromedriver WmiPrvSE conhost chrome LogonUI svchost mstsc dllhost

(50) At block **404**, the triggered discovery agent and/or the bot enforcer **242** perform a second process iteration that includes generating bot data using the process data. The block **404** can involve determining which processes in the process data from the block **402** represent bots or potential bots. In some embodiments, the triggered discovery agent can include rules or other logic that map individual processes (e.g., by name or other identifier) to particular bots or potential bots. In this way, the bot data can include, for example, a list of bots or potential bots that are executing in the computing environment. For example, the bot data can include information like the information listed below in Table 2. In certain embodiments, blocks **402** and **404** can be performed as all or part of the block **306** of FIG. **3**, such that the generated bot data represents all or part of the bot presence data discussed previously.

(51) TABLE-US-00002 TABLE 2 BOT ID BOT NAME 68878 AFTE_ICS_DemoBot 68879 AFTE_ADAM_DemoBot 68882 AFTE_ICS_Sample2 68883 AFTE_ICS_New_user_Ad_date_test 68884 AFTE_ICS_Ad_Access_Setup_UniverseAccess1

(52) At block **406**, the triggered discovery agent and/or the bot enforcer **242** execute a third process iteration that includes generating unauthorized bot data for the computing environment. The block **406** can involve determining which bots or potential bots in the bot data from the block **404** represent unauthorized bots that are present or executing in the computing environment. In general, this determination can include any of the functionality described relative to FIGS. **2** and **3**. For example, the unauthorized bot data can include information like the information listed below in Table 3. After block **406**, the process **400** ends.

(53) TABLE-US-00003 TABLE 3 MAIN MAIN PRIVATE WINDOW WINDOW MEMORY TITLE PROCESS NAME HANDLE SIZE 64 PROCESS ResumeWindow AFTE_ADAM_DemoBot 1837746 34050048 System.Diagnostics.Process (AFTE)

(54) FIG. **5** illustrates an example dashboard **500**. The dashboard **500** can be produced, for example, by the reporting module **146** of FIG. **1**. The dashboard **500** illustrates an example of bot data, such as unauthorized bot data, that can be aggregated across multiple computing environments and presented to an administrator or other user. For example, with reference to FIG. **1**, in various embodiments, a dashboard like the dashboard **500** can aggregate data, such as unauthorized bot data, for all of the computer systems **122**, or any configurable subset thereof. For example, the data, such as unauthorized bot data, can be aggregated and presented by tenant, organization, business unit, user group, customer account, combinations of the foregoing and/or the like. Dashboards such as the dashboard **500** can be generated in real-time, on demand, and/or at any desired interval.

(55) FIG. **6** illustrates an example of a computer system **600** that, in some cases, can be

representative, for example, of the central management system **140**, the RPA platforms **124**, the tenant systems **110**, the user systems **160** and/or a module or sub-component of the foregoing. The computer system **600** includes an application **622** operable to execute on computer resources **602**. The application **622** can be, for example, any of the systems or modules illustrated in FIG. **1** or FIG. **2**. In particular embodiments, the computer system **600** may perform one or more steps of one or more methods described or illustrated herein. In particular embodiments, one or more computer systems may provide functionality described or illustrated herein. In particular embodiments, encoded software running on one or more computer systems may perform one or more steps of one or more methods described or illustrated herein or provide functionality described or illustrated herein.

(56) The components of the computer system **600** may comprise any suitable physical form, configuration, number, type and/or layout. As an example, and not by way of limitation, the computer system **600** may comprise an embedded computer system, a system-on-chip (SOC), a single-board computer system (SBC) (such as, for example, a computer-on-module (COM) or system-on-module (SOM)), a desktop computer system, a laptop or notebook computer system, an interactive kiosk, a mainframe, a mesh of computer systems, a mobile telephone, a personal digital assistant (PDA), a wearable or body-borne computer, a server, or a combination of two or more of these. Where appropriate, the computer system **600** may include one or more computer systems; be unitary or distributed; span multiple locations; span multiple machines; or reside in a cloud, which may include one or more cloud components in one or more networks.

(57) In the depicted embodiment, the computer system **600** includes a processor **608**, memory **620**, storage **610**, interface **606**, and bus **604**. Although a particular computer system is depicted having a particular number of particular components in a particular arrangement, this disclosure contemplates any suitable computer system having any suitable number of any suitable components in any suitable arrangement.

(58) Processor **608** may be a microprocessor, controller, or any other suitable computing device, resource, or combination of hardware, software and/or encoded logic operable to execute, either alone or in conjunction with other components, (e.g., memory **620**), the application **622**. Such functionality may include providing various features discussed herein. In particular embodiments, processor **608** may include hardware for executing instructions, such as those making up the application **622**. As an example, and not by way of limitation, to execute instructions, processor **608** may retrieve (or fetch) instructions from an internal register, an internal cache, memory **620**, or storage **610**; decode and execute them; and then write one or more results to an internal register, an internal cache, memory **620**, or storage **610**.

(59) In particular embodiments, processor **608** may include one or more internal caches for data, instructions, or addresses. This disclosure contemplates processor **608** including any suitable number of any suitable internal caches, where appropriate. As an example, and not by way of limitation, processor **608** may include one or more instruction caches, one or more data caches, and one or more translation lookaside buffers (TLBs). Instructions in the instruction caches may be copies of instructions in memory **620** or storage **610** and the instruction caches may speed up retrieval of those instructions by processor **608**. Data in the data caches may be copies of data in memory **620** or storage **610** for instructions executing at processor **608** to operate on; the results of previous instructions executed at processor **608** for access by subsequent instructions executing at processor **608**, or for writing to memory **620**, or storage **610**; or other suitable data. The data caches may speed up read or write operations by processor **608**. The TLBs may speed up virtual-address translations for processor **608**. In particular embodiments, processor **608** may include one or more internal registers for data, instructions, or addresses. Depending on the embodiment, processor **608** may include any suitable number of any suitable internal registers, where appropriate. Where appropriate, processor **608** may include one or more arithmetic logic units (ALUs); be a multi-core processor, include one or more processors **608**; or any other suitable processor.

(60) Memory **620** may be any form of volatile or non-volatile memory including, without limitation, magnetic media, optical media, random access memory (RAM), read-only memory (ROM), flash memory, removable media, or any other suitable local or remote memory component or components. In particular embodiments, memory **620** may include random access memory (RAM). This RAM may be volatile memory, where appropriate. Where appropriate, this RAM may be dynamic RAM (DRAM) or static RAM (SRAM). Moreover, where appropriate, this RAM may be single-ported or multi-ported RAM, or any other suitable type of RAM or memory. Memory **620** may include one or more memories **620**, where appropriate. Memory **620** may store any suitable data or information utilized by the computer system **600**, including software embedded in a computer readable medium, and/or encoded logic incorporated in hardware or otherwise stored (e.g., firmware). In particular embodiments, memory **620** may include main memory for storing instructions for processor **608** to execute or data for processor **608** to operate on. In particular embodiments, one or more memory management units (MMUs) may reside between processor **608** and memory **620** and facilitate accesses to memory **620** requested by processor **608**.

(61) As an example, and not by way of limitation, the computer system **600** may load instructions from storage **610** or another source (such as, for example, another computer system) to memory **620**. Processor **608** may then load the instructions from memory **620** to an internal register or internal cache. To execute the instructions, processor **608** may retrieve the instructions from the internal register or internal cache and decode them. During or after execution of the instructions, processor **608** may write one or more results (which may be intermediate or final results) to the internal register or internal cache. Processor **608** may then write one or more of those results to memory **620**. In particular embodiments, processor **608** may execute only instructions in one or more internal registers or internal caches or in memory **620** (as opposed to storage **610** or elsewhere) and may operate only on data in one or more internal registers or internal caches or in memory **620** (as opposed to storage **610** or elsewhere).

(62) In particular embodiments, storage **610** may include mass storage for data or instructions. As an example, and not by way of limitation, storage **610** may include a hard disk drive (HDD), a floppy disk drive, flash memory, an optical disc, a magneto-optical disc, magnetic tape, or a Universal Serial Bus (USB) drive or a combination of two or more of these. Storage **610** may include removable or non-removable (or fixed) media, where appropriate. Storage **610** may be internal or external to the computer system **600**, where appropriate. In particular embodiments, storage **610** may be non-volatile, solid-state memory. In particular embodiments, storage **610** may include read-only memory (ROM). Where appropriate, this ROM may be mask-programmed ROM, programmable ROM (PROM), erasable PROM (EPROM), electrically erasable PROM (EEPROM), electrically alterable ROM (EAROM), or flash memory or a combination of two or more of these. Storage **610** may take any suitable physical form and may comprise any suitable number or type of storage. Storage **610** may include one or more storage control units facilitating communication between processor **608** and storage **610**, where appropriate.

(63) In particular embodiments, interface **606** may include hardware, encoded software, or both providing one or more interfaces for communication (such as, for example, packet-based communication) among any networks, any network devices, and/or any other computer systems. As an example, and not by way of limitation, communication interface **606** may include a network interface controller (NIC) or network adapter for communicating with an Ethernet or other wire-based network and/or a wireless NIC (WNIC) or wireless adapter for communicating with a wireless network.

(64) Depending on the embodiment, interface **606** may be any type of interface suitable for any type of network for which computer system **600** is used. As an example, and not by way of limitation, computer system **600** can include (or communicate with) an ad-hoc network, a personal area network (PAN), a local area network (LAN), a wide area network (WAN), a metropolitan area network (MAN), or one or more portions of the Internet or a combination of two or more of these.

One or more portions of one or more of these networks may be wired or wireless. As an example, computer system **600** can include (or communicate with) a wireless PAN (WPAN) (such as, for example, a BLUETOOTH WPAN), a WI-FI network, a WI-MAX network, an LTE network, an LTE-A network, a cellular telephone network (such as, for example, a Global System for Mobile Communications (GSM) network), or any other suitable wireless network or a combination of two or more of these. The computer system **600** may include any suitable interface **606** for any one or more of these networks, where appropriate.

(65) In some embodiments, interface **606** may include one or more interfaces for one or more I/O devices. One or more of these I/O devices may enable communication between a person and the computer system **600**. As an example, and not by way of limitation, an I/O device may include a keyboard, keypad, microphone, monitor, mouse, printer, scanner, speaker, still camera, stylus, tablet, touchscreen, trackball, video camera, another suitable I/O device or a combination of two or more of these. An I/O device may include one or more sensors. Particular embodiments may include any suitable type and/or number of I/O devices and any suitable type and/or number of interfaces **606** for them. Where appropriate, interface **606** may include one or more drivers enabling processor **608** to drive one or more of these I/O devices. Interface **606** may include one or more interfaces **606**, where appropriate.

(66) Bus **604** may include any combination of hardware, software embedded in a computer readable medium, and/or encoded logic incorporated in hardware or otherwise stored (e.g., firmware) to couple components of the computer system **600** to each other. As an example, and not by way of limitation, bus **604** may include an Accelerated Graphics Port (AGP) or other graphics bus, an Enhanced Industry Standard Architecture (EISA) bus, a front-side bus (FSB), a HYPERTRANSPORT (HT) interconnect, an Industry Standard Architecture (ISA) bus, an INFINIBAND interconnect, a low-pin-count (LPC) bus, a memory bus, a Micro Channel Architecture (MCA) bus, a Peripheral Component Interconnect (PCI) bus, a PCI-Express (PCI-X) bus, a serial advanced technology attachment (SATA) bus, a Video Electronics Standards Association local (VLB) bus, or any other suitable bus or a combination of two or more of these. Bus **604** may include any number, type, and/or configuration of buses **604**, where appropriate. In particular embodiments, one or more buses **604** (which may each include an address bus and a data bus) may couple processor **608** to memory **620**. Bus **604** may include one or more memory buses.

(67) Herein, reference to a computer-readable storage medium encompasses one or more tangible computer-readable storage media possessing structures. As an example, and not by way of limitation, a computer-readable storage medium may include a semiconductor-based or other integrated circuit (IC) (such, as for example, a field-programmable gate array (FPGA) or an application-specific IC (ASIC)), a hard disk, an HDD, a hybrid hard drive (HHD), an optical disc, an optical disc drive (ODD), a magneto-optical disc, a magneto-optical drive, a floppy disk, a floppy disk drive (FDD), magnetic tape, a holographic storage medium, a solid-state drive (SSD), a RAM-drive, a SECURE DIGITAL card, a SECURE DIGITAL drive, a flash memory card, a flash memory drive, or any other suitable tangible computer-readable storage medium or a combination of two or more of these, where appropriate.

(68) Particular embodiments may include one or more computer-readable storage media implementing any suitable storage. In particular embodiments, a computer-readable storage medium implements one or more portions of processor **608** (such as, for example, one or more internal registers or caches), one or more portions of memory **620**, one or more portions of storage **610**, or a combination of these, where appropriate. In particular embodiments, a computer-readable storage medium implements RAM or ROM. In particular embodiments, a computer-readable storage medium implements volatile or persistent memory. In particular embodiments, one or more computer-readable storage media embody encoded software.

(69) Herein, reference to encoded software may encompass one or more applications, bytecode, one or more computer programs, one or more executables, one or more instructions, logic, machine

code, one or more scripts, or source code, and vice versa, where appropriate, that have been stored or encoded in a computer-readable storage medium. In particular embodiments, encoded software includes one or more application programming interfaces (APIs) stored or encoded in a computer-readable storage medium. Particular embodiments may use any suitable encoded software written or otherwise expressed in any suitable programming language or combination of programming languages stored or encoded in any suitable type or number of computer-readable storage media. In particular embodiments, encoded software may be expressed as source code or object code. In particular embodiments, encoded software is expressed in a higher-level programming language, such as, for example, C, Perl, or a suitable extension thereof. In particular embodiments, encoded software is expressed in a lower-level programming language, such as assembly language (or machine code). In particular embodiments, encoded software is expressed in JAVA. In particular embodiments, encoded software is expressed in Hyper Text Markup Language (HTML), Extensible Markup Language (XML), or other suitable markup language.

(70) Depending on the embodiment, certain acts, events, or functions of any of the algorithms described herein can be performed in a different sequence, can be added, merged, or left out altogether (e.g., not all described acts or events are necessary for the practice of the algorithms). Moreover, in certain embodiments, acts or events can be performed concurrently, e.g., through multi-threaded processing, interrupt processing, or multiple processors or processor cores or on other parallel architectures, rather than sequentially. Although certain computer-implemented tasks are described as being performed by a particular entity, other embodiments, are possible in which these tasks are performed by a different entity.

(71) Conditional language used herein, such as, among others, “can,” “might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or states. Thus, such conditional language is not generally intended to imply that features, elements and/or states are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or states are included or are to be performed in any particular embodiment.

(72) While the above detailed description has shown, described, and pointed out novel features as applied to various embodiments, it will be understood that various omissions, substitutions, and changes in the form and details of the devices or algorithms illustrated can be made without departing from the spirit of the disclosure. As will be recognized, the processes described herein can be embodied within a form that does not provide all of the features and benefits set forth herein, as some features can be used or practiced separately from others. The scope of protection is defined by the appended claims rather than by the foregoing description. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

Claims

1. A method of automatic discovery and control of robotic process automation (RPA) bots, the method comprising, by a computer system: developing, via an RPA bot discovery agent executing in a computing environment, process data comprising a list of processes executing in the computing environment; identifying, via the RPA bot discovery agent, at least one RPA bot executing in the computing environment by determining that one or more processes in the list of processes represent the at least one RPA bot; responsive to the identifying, generating, via the RPA bot discovery agent, RPA bot presence data for the computing environment, the generated RPA bot presence data identifying the at least one RPA bot and an RPA platform of the at least one RPA bot; determining, using the RPA bot presence data, that the at least one RPA bot is unauthorized; responsive to the determining that the at least one RPA bot is unauthorized, automatically

identifying one or more deficiencies in the at least one RPA bot that resulted in the determination that the at least one RPA bot is unauthorized; and automatically correcting the one or more deficiencies in the at least one RPA bot.

2. The method of claim 1, comprising triggering RPA bot discovery by a plurality of RPA bot discovery agents executing in a plurality of computing environments, wherein the RPA bot presence data is received responsive to the triggering.

3. The method of claim 2, wherein the triggering comprises sending, to each of the plurality of RPA bot discovery agents, a command to execute the RPA bot discovery upon receipt of the command.

4. The method of claim 2, wherein the triggering comprises sending, to each of the plurality of RPA bot discovery agents, a command to schedule the RPA bot discovery.

5. The method of claim 1, wherein the determining that the at least one RPA bot is unauthorized comprises mapping the RPA bot presence data to a plurality of signatures.

6. The method of claim 5, wherein: the plurality of signatures comprise one or more signatures of authorized RPA bots; and the determination that the at least one RPA bot is unauthorized is based on a determination that the RPA bot presence data does not match any of the one or more signatures of authorized RPA bots.

7. The method of claim 5, wherein: the plurality of signatures comprise one or more signatures of unauthorized RPA bots; and the determination that the at least one RPA bot is unauthorized is based on a determination that the RPA bot presence data matches at least one of the one or more signatures of unauthorized RPA bots.

8. The method of claim 1, wherein the determining that the at least one RPA bot is unauthorized comprises executing a heuristics analysis using the RPA bot presence data, the heuristics analysis comprising matching the RPA bot presence data to an identified unauthorized RPA bot.

9. The method of claim 1, wherein the RPA bot presence data indicates an instruction set of at least one bot.

10. The method of claim 1, wherein the RPA bot presence data indicates an observed behavior of at least one RPA bot.

11. The method of claim 1, comprising: deploying a plurality of RPA bot discovery agents in a plurality of computing environments; triggering bot discovery by the plurality of RPA bot discovery agents in the plurality of computing environments; receiving RPA bot presence data from each of the plurality of RPA bot discovery agents; determining, based on the RPA bot presence data received from each of the plurality of bot discovery agents, whether an unauthorized RPA bot is present in any of the plurality of computing environments; and responsive to a determination, based on the RPA bot presence data received from each of the plurality of RPA bot discovery agents, that at least one unauthorized bot is present in the plurality of computing environments, automatically initiating a remediation workflow relative to the at least one unauthorized bot.

12. The method of claim 1, wherein: the determining that the at least one RPA bot is unauthorized is based on the RPA platform being a non-approved RPA platform; the identifying the one or more deficiencies comprises identifying that the at least one RPA bot is deployed using the non-approved RPA platform; and the automatically correcting comprises automatically stopping the at least one RPA bot and redeploying the at least one RPA bot using an approved RPA platform.

13. The method of claim 1, wherein the identifying of the at least one RPA bot is based on a mapping of process names in the list of processes to particular RPA bots.

14. The method of claim 1, wherein the RPA bot discovery agent continuously executes a bot discovery process in the computing environment.

15. The method of claim 1, wherein: the RPA bot presence data comprising an instruction set of the at least one RPA bot; the determining that the at least one RPA bot is unauthorized is based on the instruction set including an unauthorized instruction; the identifying the one or more deficiencies comprises identifying the unauthorized instruction; and the automatically correcting comprises: stopping the at least one RPA bot; modifying the instruction set to correct the unauthorized

instruction; and starting the at least one RPA bot with the modified instruction set.

16. The method of claim 15, wherein the modifying comprises removing the unauthorized instruction from the instruction set.

17. The method of claim 1, comprises incrementally learning new settings for the determining, based on the RPA bot presence data, that the at least one RPA bot is unauthorized.

18. The method of claim 1, comprising receiving new settings for the determining, based on the RPA bot presence data, that the at least one RPA bot is unauthorized.

19. A computer system comprising a processor and memory, wherein the processor and the memory in combination are operable to implement a method of automatic discovery and control of robotic process automation (RPA) bots, the method comprising: developing, via an RPA bot discovery agent executing in a computing environment, process data comprising a list of processes executing in the computing environment; identifying, via the RPA bot discovery agent, at least one RPA bot executing in the computing environment by determining that one or more processes in the list of processes represent the at least one RPA bot; responsive to the identifying, generating, via the RPA bot discovery agent, RPA bot presence data for the computing environment, the generated RPA bot presence data identifying the at least one RPA bot and an RPA platform of the at least one RPA bot; determining, using the RPA bot presence data, that the at least one RPA bot is unauthorized; responsive to the determining that the at least one RPA bot is unauthorized, automatically identifying one or more deficiencies in the at least one RPA bot that resulted in the determination that the at least one RPA bot is unauthorized; and automatically correcting the one or more deficiencies in the at least one RPA bot.

20. A computer-program product comprising a non-transitory computer-usable medium having computer-readable program code embodied therein, the computer-readable program code adapted to be executed to implement a method of automatic discovery and control of robotic process automation (RPA) bots, the method comprising: developing, via an RPA bot discovery agent executing in a computing environment, process data comprising a list of processes executing in the computing environment; identifying, via the RPA bot discovery agent, at least one RPA bot executing in the computing environment by determining that one or more processes in the list of processes represent the at least one RPA bot; responsive to the identifying, generating, via the RPA bot discovery agent, RPA bot presence data for the computing environment, the generated RPA bot presence data identifying the at least one RPA bot and an RPA platform of the at least one RPA bot; determining, using the RPA bot presence data, that the at least one RPA bot is unauthorized; responsive to the determining that the at least one RPA bot is unauthorized, automatically identifying one or more deficiencies in the at least one RPA bot that resulted in the determination that the at least one RPA bot is unauthorized; and automatically correcting the one or more deficiencies in the at least one RPA bot.
